What Are You Searching For? A Remote Keylogging Attack on Search Engine Autocomplete

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- Published: date not stated
- Original: <https://www.usenix.org/conference/usenixsecurity19/presentation/monaco>
- Preserved from: https://www.usenix.org/conference/usenixsecurity19/presentation/monaco (stored) on 2026-08-11
- Capture timestamp: 20191114161052
- Licence: unknown

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Abstract:

Many search engines have an autocomplete feature that presents a list of suggested queries to the user as they type. Autocomplete induces network traffic from the client upon changes to the query in a web page. We describe a remote keylogging attack on search engine autocomplete. The attack integrates information leaked by three independent sources: the timing of keystrokes manifested in packet inter-arrival times, percent-encoded Space characters in a URL, and the static Huffman code used in HTTP2 header compression. While each source is a relatively weak predictor in its own right, combined, and by leveraging the relatively low entropy of English language, up to 15% of search queries are identified among a list of 50 hypothesis queries generated from a dictionary with over 12k words. The attack succeeds despite network traffic being encrypted. We demonstrate the attack on two popular search engines and discuss some countermeasures to mitigate attack success.

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BibTeX

@inproceedings {236286, author = {John V. Monaco}, title = {What Are You Searching For? A Remote Keylogging Attack on Search Engine Autocomplete}, booktitle = {28th {USENIX} Security Symposium ({USENIX} Security 19)}, year = {2019}, isbn = {978-1-939133-06-9}, address = {Santa Clara, CA}, pages = {959--976}, url = {https://www.usenix.org/conference/usenixsecurity19/presentation/monaco}, publisher = {{USENIX} Association}, month = aug, }

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